

Assignment 1(a)

Title: Write a program to implement Parallel BFS using OpenMP

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```
#include <iostream>
#include <queue>
#include <omp.h>

using namespace std;

/* Node structure */
class node
{
public:
    int data;
    node *left;
    node *right;

    node(int val)
    {
        data = val;
        left = right = NULL;
    }
};

/* BFS Class */
class Breadthfs
{
public:
    node* insert(node*, int);
    void bfs(node*);
};

/* Insert node in level order */
node* Breadthfs::insert(node *root, int data)
{
    if (!root)
        return new node(data);

    queue<node*> q;
    q.push(root);

    while (!q.empty())
```

```

{
    node *temp = q.front();
    q.pop();

    if (!temp->left)
    {
        temp->left = new node(data);
        return root;
    }
    else
        q.push(temp->left);

    if (!temp->right)
    {
        temp->right = new node(data);
        return root;
    }
    else
        q.push(temp->right);
}

return root;
}

/* Parallel BFS traversal */
void Breadthfs::bfs(node *head)
{
    if (!head) return;

    queue<node*> q;
    q.push(head);

    while (!q.empty())
    {
        int qSize = q.size();

        #pragma omp parallel for
        for (int i = 0; i < qSize; i++)
        {
            node *currNode;

            #pragma omp critical
            {
                currNode = q.front();
                q.pop();
                cout << currNode->data << " ";
            }
        }
    }
}

```

```

        #pragma omp critical
        {
            if (currNode->left)
                q.push(currNode->left);

            if (currNode->right)
                q.push(currNode->right);
        }
    }
}

/* Main function */
int main()
{
    Breadthfs tree;
    node *root = NULL;

    int data;
    char ans;

    do
    {
        cout << "Enter data: ";
        cin >> data;

        root = tree.insert(root, data);

        cout << "Insert more nodes? (y/n): ";
        cin >> ans;

    } while (ans == 'y' || ans == 'Y');

    cout << "\nParallel BFS Traversal:\n";
    tree.bfs(root);

    return 0;
}

```

OUTPUT:

```
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
datanalytics1@datanalytics1-OptiPlex-7050:~$ g++ -fopenmp parallel_bfs.cpp -o p
arallel_bfs
datanalytics1@datanalytics1-OptiPlex-7050:~$ ./parallel_bfs
Enter data => 1
Do you want to insert one more node? (y/n): y
Enter data => 2
Do you want to insert one more node? (y/n): y
Enter data => 3
Do you want to insert one more node? (y/n): y
Enter data => 4
Do you want to insert one more node? (y/n): y
Enter data => 5
Do you want to insert one more node? (y/n): n

Parallel BFS Traversal:
1
2      3
4      5
```